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SECTION:	S-8
SUBJECT:	FULL STACK APPLICATION DEVELOPMENT

#SKILL - 7 REST API - CRUD Operations using Response Entity

Tasks:

1. **Create a Course entity with fields: courseId, title, duration, fee.**

   <u>Output:-</u>

LibraryController.java | LibraryAppApplication.java | Course.java

```java
package com.klu.course.entity;

import jakarta.persistence.Entity;

@Entity
public class Course {

    @Id
    private int courseId;
    private String title;
    private String duration;
    private double fee;

    public Course() {}

    public int getCourseId() {
        return courseId;
    }

    public void setCourseId(int courseId) {
        this.courseId = courseId;
    }

    public String getTitle() {
        return title;
    }

    public void setTitle(String title) {
        this.title = title;
    }

    public String getDuration() {
        return duration;
    }

    public void setDuration(String duration) {
```

2. **Create a Service layer that performs full CRUD operations.**

<u>Output:-</u>

LibraryController.java | LibraryAppApplication.java | Course.java | CourseService.java

```java
package com.klu.course.service;

import java.util.List;

@Service
public class CourseService {

    @Autowired
    private CourseRepository repo;

    public Course addCourse(Course course) {
        return repo.save(course);
    }

    public List<Course> getAllCourses() {
        return repo.findAll();
    }

    public Course updateCourse(Course course) {
        return repo.save(course);
    }

    public void deleteCourse(int id) {
        repo.deleteById(id);
    }

    public List<Course> searchByTitle(String title) {
        return repo.findByTitle(title);
    }
}
```

3. **Implement REST endpoints using @PostMapping, @PutMapping, @DeleteMapping, and @GetMapping.**

Output:-

LibraryAppAp... | Course.java | CourseServic... | CourseReposi... | CourseContro...

```java
package com.klu.course.controller;

import java.util.List;

@RestController
@RequestMapping("/courses")
public class CourseController {

    @Autowired
    private CourseService service;

    @PostMapping
    public ResponseEntity<?> addCourse(@RequestBody Course course) {
        Course c = service.addCourse(course);
        return new ResponseEntity<>(c, HttpStatus.CREATED);
    }

    @GetMapping
    public ResponseEntity<?> getAllCourses() {
        List<Course> list = service.getAllCourses();
        return new ResponseEntity<>(list, HttpStatus.OK);
    }

    @PutMapping
    public ResponseEntity<?> updateCourse(@RequestBody Course course) {
        Course c = service.updateCourse(course);
        return new ResponseEntity<>(c, HttpStatus.OK);
    }

    @DeleteMapping("/{id}")
    public ResponseEntity<?> deleteCourse(@PathVariable int id) {
        service.deleteCourse(id);
        return new ResponseEntity<>("Course Deleted Successfully", HttpStatus.OK);
    }

    @GetMapping("/search/{title}")
    public ResponseEntity<?> searchCourse(@PathVariable String title) {
        List<Course> list = service.searchByTitle(title);

        if(list.isEmpty()) {
            return new ResponseEntity<>("Course Not Found", HttpStatus.NOT_FOUND);
        }

        return new ResponseEntity<>(list, HttpStatus.OK);
    }
}
```

4 . Use ResponseEntity to return:

a. Success messages

b. Error messages

c. Status codes such as OK, CREATED, NOT_FOUND, BAD_REQUEST

Output:-

```java
package com.klu.course.Repository;

import java.util.List;

public interface CourseRepository extends JpaRepository<Course, Integer> {

    List<Course> findByTitle(String title);

}
```

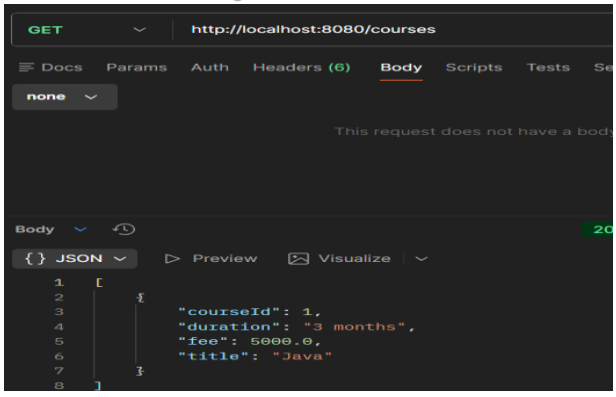
5. Add a search endpoint /courses/search/{title} to filter courses by title.

Output:-

```java
package com.klu.course.Repository;

import java.util.List;

public interface CourseRepository extends JpaRepository<Course, Integer> {

    List<Course> findByTitle(String title);

}
```

6. Test all endpoints (valid and invalid cases) using Postman.

Output:-



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http://localhost:8080/courses

POST http://localhost:8080/courses

Docs Params Auth Headers (8) Body Scripts

raw JSON

```
{
  "courseId": 1,
  "title": "Java",
  "duration": "3 months",
  "fee": 5000
}
```

Body

JSON Preview Visualize

```
{
    "courseId": 1,
    "duration": "3 months",
    "fee": 5000.0,
    "title": "Java"
}
```

PUT http://localhost:8080/courses

Docs Params Auth Headers (8) Body Scripts

raw JSON

```
{
  "courseId": 1,
  "title": "Advanced Java",
  "duration": "4 months",
  "fee": 7000
}
```

Body

JSON Preview Visualize

```
{
    "courseId": 1,
    "duration": "4 months",
    "fee": 7000.0,
    "title": "Advanced Java"
}
```

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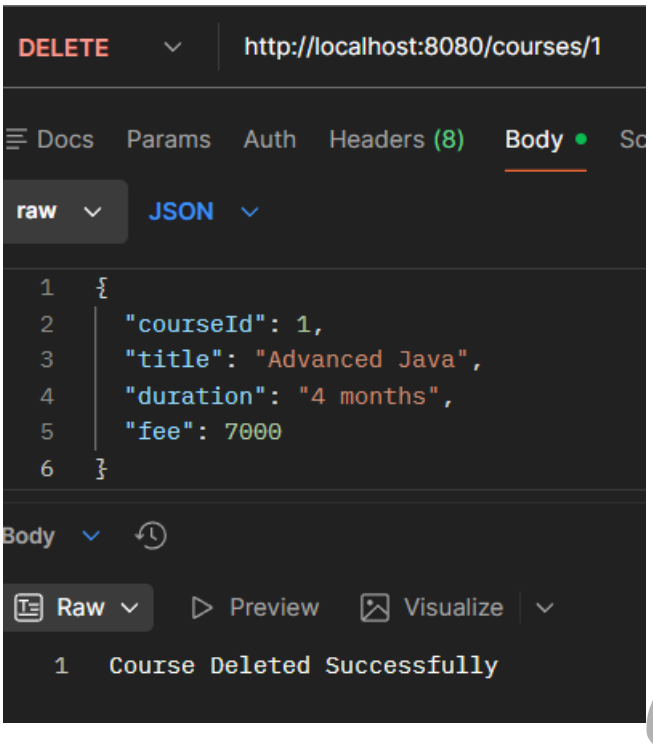
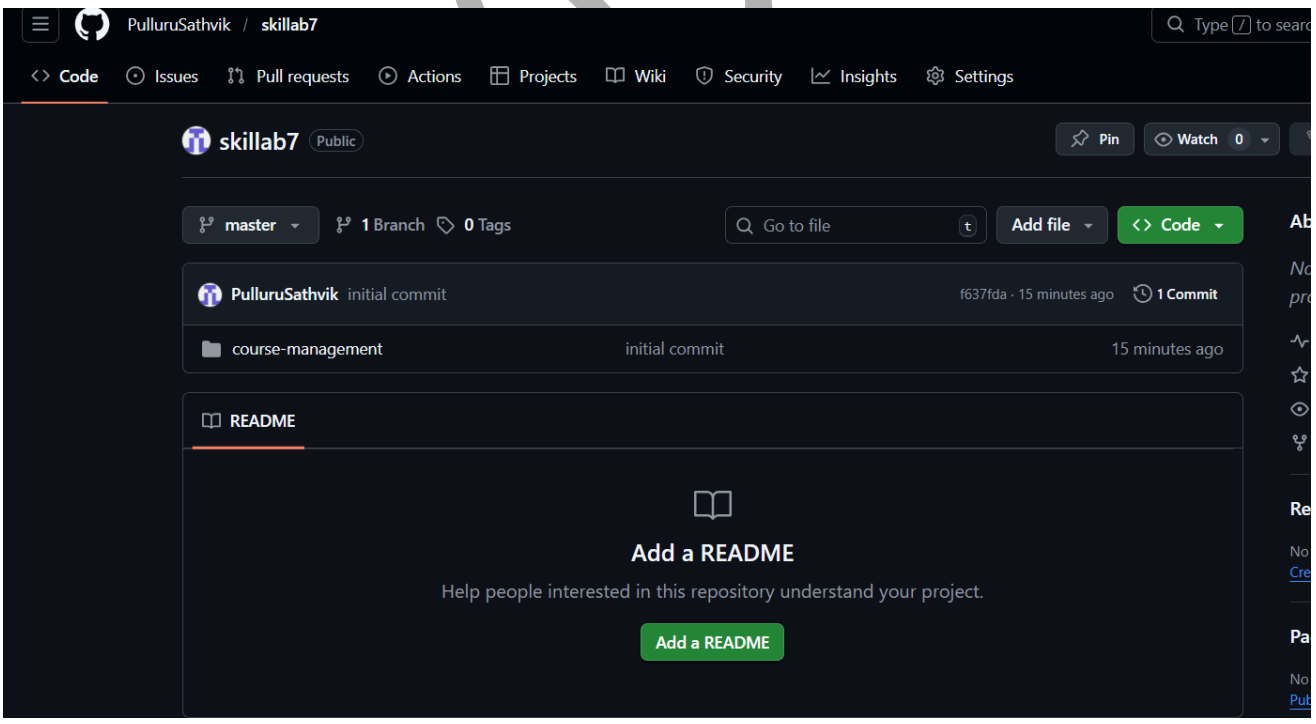


7. Push the Spring Boot project to GitHub.

Output:-



GitHub Repo:- https://github.com/PulluruSathvik/skillab7

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